CSCI 460 - Programming Assignment I

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GitHub Link: https://github.com/Loubyu/CSCI460F2026_Programing_Assignment1

Dataset and cleaning

For this assignment I used the UCI Bank Marketing dataset (bank-full.csv). It has 45,211 rows and 17 columns, with 16 columns being used to predict whether someone said yes or no to the marketing offer. I loaded it with pandas, removed the extra quotation marks, and changed the categories into numbers starting at 0 so the model could work with them. I also used dropna(), but it turns out that did not remove any rows.

The biggest thing I ran into was the difference between an actual NA and a value that literally says "unknown." There were 288 unknown jobs, 1,857 unknown education values, 13,020 unknown contact values, and 36,959 unknown values for poutcome. At first I thought I was missing something because dropna() was not doing anything. After talking with Dr. Klement, I understood that this was part of the point of the cleaning process: an unknown category is not automatically the same thing as a missing NA. I kept those values instead of throwing out a huge part of the dataset.

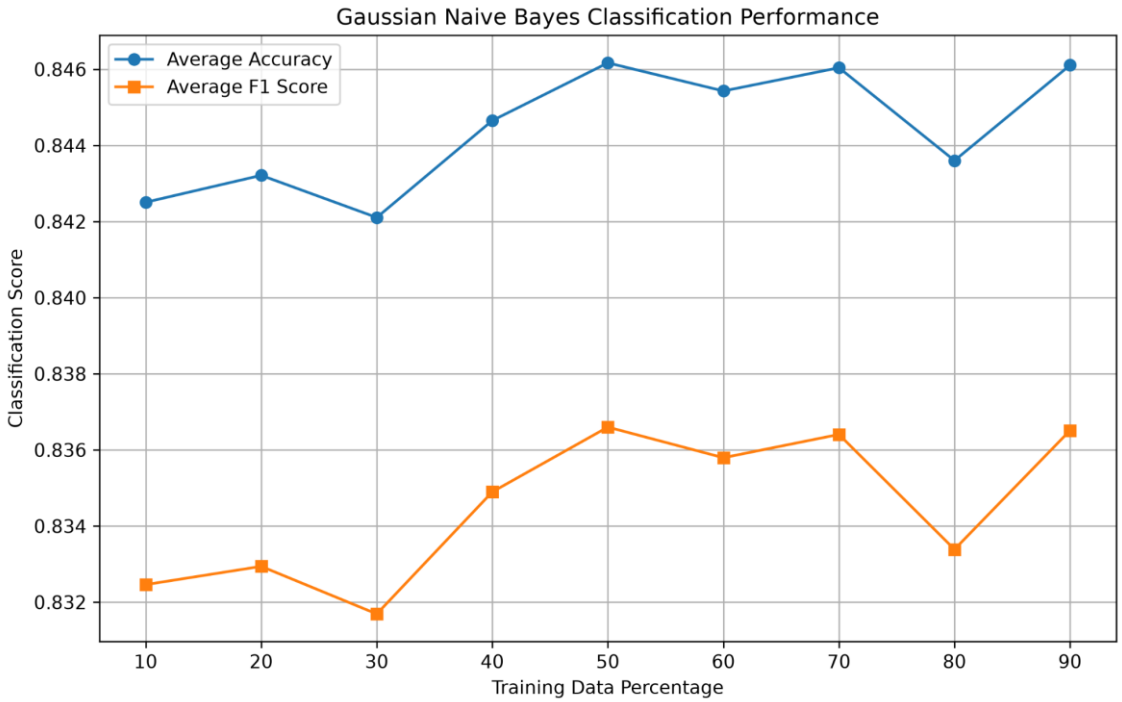
What I tested

I wanted to see whether giving Gaussian Naive Bayes more training data would actually improve its accuracy and F1 score. I tested training sizes from 10% to 90%, increasing by 10% each time. For every size I ran 10 random trials using stratified splits, so the yes/no proportions stayed similar. That gave me 90 trials total. I calculated the mean and standard deviation for accuracy and weighted F1. The individual trials are in trial_results.csv on GitHub.

Results

Train	Avg. accuracy	Accuracy SD	Avg. F1	F1 SD
10%	0.8425	0.0068	0.8325	0.0089
20%	0.8432	0.0033	0.8329	0.0044
30%	0.8421	0.0036	0.8317	0.0044
40%	0.8447	0.0026	0.8349	0.0032
50%	0.8462	0.0027	0.8366	0.0032
60%	0.8454	0.0029	0.8358	0.0033
70%	0.8460	0.0045	0.8364	0.0055
80%	0.8436	0.0028	0.8334	0.0033
90%	0.8461	0.0041	0.8365	0.0049

The scores stayed pretty close together. Average accuracy was about 84.2% to 84.6%, and average F1 was about 83.2% to 83.7%. The 50% split came out on top for both in this run, which was not what I expected. I thought more training data would make the scores steadily go up, but that is not what happened.



What I got out of it

I do not think the 50% split being the highest means that 50% is somehow the best amount of training data. The differences were small, and each split used different random training and testing rows. The standard deviations help show that. The 10% training split had the most variation (0.0068 for accuracy and 0.0089 for F1), which makes sense since it was working with the smallest training set.

There are also some limits to the model itself. Gaussian Naive Bayes makes assumptions about how the features behave, and some of those assumptions may not fit this dataset very well. I also had to turn categories like job and contact into numbers, even though those numbers do not really mean one category is bigger than another. On top of that, there are a lot of unknown values. Accuracy by itself can also hide how well the model handles the less common yes responses.

Overall, I was expecting a clearer improvement when the training percentage increased. Instead, the results were pretty consistent across the board. My takeaway is that more training data does not automatically fix the limitations of the model or how the data is represented.